

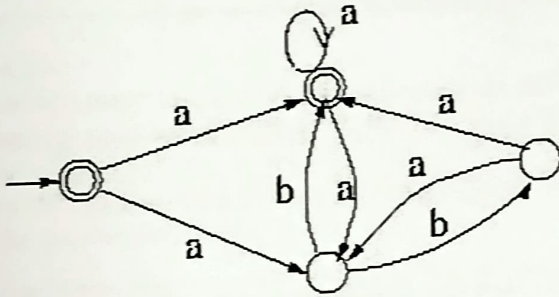
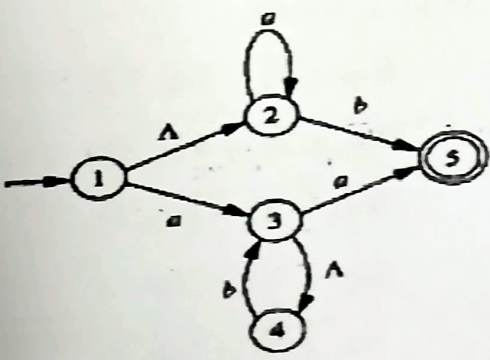
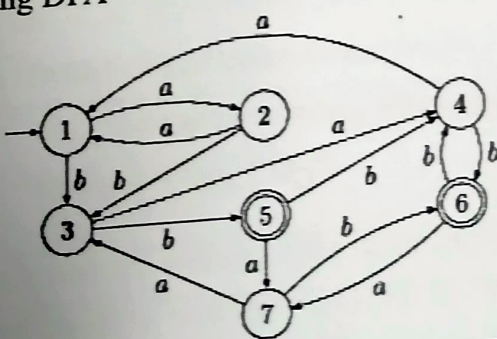
Name of Student: _____

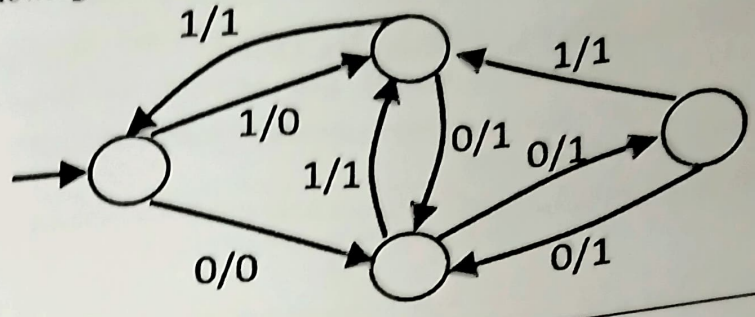
Scholar No.: _____

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL**Department of Computer Science & Engineering (CSE)****Supplementary Examination Dec - 2025**Subject Name: **Formal Language and Automata Theory**
MM:60Subject Code: **CSE-225**
Time Allotted: **180 min**

NOTE: 1. All questions are compulsory.

2. Mark answer number clearly.

1.A	Construct DFA that will accepts all the strings of $\{0, 1\}$ in which every '0' is not followed by at least two 1's.	5
B.	Find the regular expression 	5
2.A	Find DFA equivalent to the following NFA. 	5
B.	Minimize the following DFA 	5

3.A	The languages accepted by the following Grammar $G(\{S\}, \{a, b\}, S, P)$ and prove that grammar G is ambiguous. $S \rightarrow aSb / bSa / \epsilon$	5
B.	Convert the following Mealy machine to Moore machine. 	5
4.A	Context Free Grammar for the following languages. (i) $L = \{ww^r \mid w \in (0+1)^* \text{ and } w^r \text{ is reverse of } w\}$ (ii) $L = \{a^n b^m c^n \mid m, n > 0\}$	5
B.	Prove that (i) CFGs are closed under Kleen's closure operation. (ii) CFGs are not closed under intersection operation.	5
5 A	Find LBA for the language $L = \{n_a(w) = n_b(w) \mid w \in \{a,b\}^*\}$	5
B	Construct Context Free Grammar for the following PDA $(\{q_0, q_1, q_2\}, \{a, b, c\}, q_0, \{q_2\}, \delta, \{A, Z\})$ The transitions δ of PDA are as follows: $\delta(q_0, a, Z) \rightarrow \delta(q_0, XZ)$ $\delta(q_0, a, X) \rightarrow \delta(q_0, XX)$ $\delta(q_0, b, X) \rightarrow \delta(q_1, \lambda)$ $\delta(q_1, b, X) \rightarrow \delta(q_1, \lambda)$ $\delta(q_1, c, Z) \rightarrow \delta(q_2, Z)$	5
6.A	Construct PDA for the following Context Free Language. $L = \{n_a(w) \neq 2n_b(w) \mid w \in (a+b)^*\}$	5
B	Design a Turing Machine for given language L. $L = \{a^n b^n a^n \mid n > 0\}$	5